

QUERY PROCESSING

Question 1: E-Commerce Customer Transactions Dataset

CustomerID	CustomerName	Gender	Age	ProductCategory	Quantity	UnitPrice	PurchaseDate	City
C101	Raj Kumar	Male	28	Electronics	2	15000 INR	10-01-2025	Chennai
C102	Priya S	Female	35	Fashion	3	1200 INR	11-01-2025	chennai
C103	Arun V	Male	150	Electronics	1	300 USD	12-01-2025	CHENNAI
C104	Kavya R	Female	0	Grocery	-2	800 INR	15-01-2025	Coimbatore
C105	Rahul P	Male	42	Fashion	5	2500 INR	18-01-2025	Madurai
C105	Rahul P	Male	42	Fashion	5	2500 INR	18-01-2025	Madurai
NULL	Sneha M	Female	30	Electronics	2	450 USD	05-02-2025	Trichy
C107	Deepak N	Male		Grocery	3	600 INR	07-02-2025	Tiruchirappalli
C108	Meena K	Female	27	Fashion	1	1800 INR	2025-13-01	Salem
C109	John D	Male	31	Electronics	4	250 EUR	15-02-2025	Chennai

Data Issues:

- Missing Customer IDs
- Duplicate transactions
- Negative quantities
- Unit prices recorded in different currencies
- Age values such as 0, 150, and blanks
- Inconsistent city names (e.g., Chennai, chennai, CHENNAI)
- Incorrect purchase dates

Tasks:

1. Identify and classify all data quality issues.

Answer:- The dataset contains missing IDs, duplicate transactions, negative quantities, mixed currencies, invalid ages, inconsistent city names, and an invalid purchase date.

Issue	Example	Classification
Missing Customer ID	Sneha M has NULL	Missing value
Duplicate transaction	C105 appears twice	Duplicate
Negative quantity	C104 has -2	Invalid value
Different currencies	INR, USD, EUR	Inconsistency
Invalid age	Age 0	Invalid value
Missing age	Deepak N has no age	Missing value
Unrealistic age	Age 150 if present in the intended dataset specification	Outlier
Inconsistent city names	Chennai, chennai, CHENNAI	Formatting inconsistency
City spelling variation	Trichy/Tiruchirappalli	Standardization issue
Invalid date	2025-13-01	Invalid format/value
Mixed date formats	10-01-2025 and 2025-13-01	Formatting inconsistency

2. Design a cleaning strategy for each issue.

Answer:- **2. Cleaning strategy for each issue**

1. Missing Customer ID

- Do not guess an ID without supporting information.
- Assign a temporary unique identifier such as UNKNOWN_001.
- If the original source can be checked, recover the correct ID.
- Maintain a data-quality flag.

2. Duplicate transactions

- Compare Customer ID, customer name, product, quantity, price and purchase date.
- The two C105 records are identical, so retain only one.

3. Negative quantity

- Quantity cannot normally be negative for a sales transaction.
- Check whether it represents a return.
- If it is an accidental entry, correct or remove it.
- If it represents a return, store it separately as a return transaction.

4. Different currencies

- Convert all monetary values into a common reporting currency, such as INR.
- Use the appropriate historical exchange rate for the purchase date.
- Store both the original amount/currency and converted amount.

5. Invalid ages

- Replace impossible values such as 0 or 150 with NULL.
- Impute only when justified by reliable demographic information.

- Otherwise retain as missing.
- 6. **City names**
 - Convert all city names to a standard format.
 - Example:
 - Chennai
 - chennai
 - CHENNAI

→ **Chennai**
 - Map equivalent names such as Trichy and Tiruchirappalli to one approved name.
- 7. **Invalid dates**
 - Convert all dates to ISO format: YYYY-MM-DD.
 - Validate month/day combinations.
 - 2025-13-01 is invalid because month 13 does not exist.
 - Check the original source before correcting it.

3. Perform data standardization and transformation.

Answer:- A standardized dataset could use:

Field	Standardization
Customer ID	Unique alphanumeric ID
Name	Title case
Gender	Male/Female/Other/Unknown
Age	Integer within valid range
Category	Standard category names
Quantity	Positive integer
Unit Price	Numeric
Currency	Standard ISO currency code
Purchase Date	YYYY-MM-DD
City	Standard city name
Revenue	Quantity × Unit Price after currency conversion

4. Analyze how cleaning affects total revenue calculations.

Answer:- Before cleaning, the arithmetic total of $\text{Quantity} \times \text{Unit Price}$ is:

₹62,800 + 300 USD + 1,000 EUR, because the dataset mixes currencies. Therefore, **₹62,800 cannot legitimately be treated as one total revenue figure**.

There are also two major transaction problems:

- Duplicate C105 contributes an extra **₹12,500**.
- C104 has $-2 \times ₹800 = -₹1,600$, which should not be treated as a normal sale.

After removing the duplicate and invalid negative sale, the raw numerical total becomes **51,900**, but this is still **not a valid consolidated revenue figure** because USD and EUR values remain unconverted.

Therefore, the correct procedure is:

Cleaned Revenue = $\Sigma(\text{Valid Quantity} \times \text{Unit Price} \times \text{Exchange Rate})$

The assessment does not provide exchange rates, so an exact final consolidated revenue cannot be calculated from the supplied data alone.

5. Generate a report discussing business risks if the data were not cleaned.

Answer:- Poor-quality transaction data can cause:

- Incorrect revenue reporting
- Wrong customer segmentation
- Incorrect product performance rankings
- Financial reporting errors
- Incorrect sales forecasts
- Duplicate revenue counting
- Incorrect regional analysis
- Poor inventory decisions
- Loss of customer trust
- Incorrect management decisions

Conclusion: Data cleaning is essential before using the dataset for customer segmentation, revenue analysis or business intelligence.

Question 2: Hospital Patient Records Dataset

PatientID	Name	Age	Gender	AdmissionDate	DischargeDate	Diagnosis	TreatmentCost	DoctorID
P1001	Ravi Kumar	45	M	10-01-2025	15-01-2025	Diabetes	25000	D101
P1002	Priya S	38	F	12-01-2025	11-01-2025	Hypertension	18000	D102
P1003	Arun V	125	M	13-01-2025	20-01-2025	diabetes	22000	D101
NULL	Kavya R	29	F	15-01-2025	17-01-2025	Diabtes	15000	D103
P1005	Rahul p	51	M	18-01-2025	25-01-2025	Asthma	NULL	D104
P1005	Rahul p	51	M	18-01-2025	25-01-2025	Asthma	NULL	D104
P1006	Sneha M	34	F	15-01-2025	18-01-2025	Hypertension	12000	D102
P1007	Deepa k N		M	01-02-2025	05-02-2025	Asthma	10000	D104
P1008	Meena K	26	F	10-02-2025	15-02-2025	HYPERTENSION	13500	D102
P1009	John D	40	M	12-02-2025	18-02-2025	Diabetes	20000	D101

Data Issues:-

- Missing patient IDs
- Duplicate patient records
- Discharge dates before admission dates
- Null treatment costs
- Inconsistent diagnosis names
- Multiple formats of dates
- Unrealistic patient ages (>120)

Tasks:-

1. Detect all inconsistencies and anomalies.

Answer:-

Issue	Example	Type
Missing Patient ID	Kavya R	Missing value
Duplicate record	P1005 appears twice	Duplicate
Discharge before admission	P1002: 12-Jan admission, 11-Jan discharge	Logical error
Missing treatment cost	P1005	Missing value
Diagnosis spelling	Diabtes	Typographical error
Diagnosis case	diabetes, Diabetes	Formatting issue

Issue	Example	Type
Diagnosis case	HYPERTENSION	Formatting issue
Unrealistic age	Arun V = 125	Outlier
Date formatting	Different date formats	Formatting issue

2. Develop rules for handling missing and invalid values.

Answer:- **Patient ID**

- Search the original hospital record.
- If unavailable, assign a temporary unique identifier.
- Never assign an ID already belonging to another patient.

Treatment cost

- Check the billing department.
- If the value cannot be recovered, use an appropriate statistical imputation only for analytical purposes.
- Mark the value as imputed.

Age

- Validate age against a reasonable range.
- Age 125 should be flagged.
- Verify against the patient's original record.
- Do not automatically delete the patient.

Dates

- Convert all dates to YYYY-MM-DD.
- Ensure:

Discharge Date $\geq$ Admission Date

If this condition fails, send the record for verification.

Duplicate records

P1005 appears twice with identical information.

Retain one record and remove the duplicate.

3. Create a process to standardize diagnosis names.

Answer:- Use a controlled vocabulary:

Original	Standardized
diabetes	Diabetes
Diabetes	Diabetes
Diabtes	Diabetes
hypertension	Hypertension
HYPERTENSION	Hypertension
Asthma	Asthma

4. Evaluate how cleaning changes average treatment cost and hospital stay duration.

Answer:- Known treatment costs are:

$$25,000 + 18,000 + 22,000 + 15,000 + 12,000 + 13,500 + 20,000 \\ = \text{₹}125,500$$

There are 7 records with known treatment costs.

Therefore:

$$\text{Average treatment cost} = \text{₹}125,500 / 7 = \text{₹}17,928.57$$

The duplicate P1005 has a NULL treatment cost, so removing it does not change this calculation.

Hospital stay

The calculated stays include:

- P1001 = 5 days
- P1002 = **-1 day — invalid**
- P1003 = 7 days
- P1004 = 2 days
- P1005 = 7 days
- P1006 = 3 days
- P1007 = 4 days
- P1008 = 5 days
- P1009 = 6 days

The duplicate P1005 should be removed.

If the invalid P1002 stay is excluded, the average of the valid stays is approximately:

4.875 days

This demonstrates that cleaning prevents an impossible negative hospital stay from distorting hospital performance analysis.

5. Recommend data governance measures to prevent future errors.

Answer:- The hospital should implement:

1. Mandatory Patient ID fields
2. Unique constraints on Patient IDs
3. Date validation rules
4. Automated admission/discharge validation
5. Controlled diagnosis vocabulary
6. Age-range validation
7. Mandatory treatment-cost checks
8. Duplicate detection
9. Audit logs
10. Role-based access control
11. Regular data-quality reports
12. Staff training

Conclusion: Strong governance prevents incorrect patient information from affecting clinical and financial decisions.

Question 3: Banking Loan Approval Dataset

CustomerID	AnnualIncome	CreditScore	LoanAmount	EmploymentStatus	Age	ExistingDebt	LoanStatus
L001	750000	720	300000	Full Time	35	50000	Approved
L002	900K	680	400000	FT	42	100000	Approved
L003	1200000	NULL	600000	fulltime	28	200000	Approved
L004	150000	450	-200000	Part Time	30	50000	Rejected
L005	5000000	790	1000000	Self Employed	16	10000	Approved
L006	450000	350	250000	Unemployed	40	300000	Rejected
L007	75000000	800	500000	Full Time	45	100000	Approved
L008	650000	690	350000	FT	33	70000	Approved
L008	650000	690	350000	FT	33	70000	Approved
L010	NULL	710	300000	Full Time	37	80000	Approved

Data Issues:-

- Missing credit scores
- Income values recorded in different units
- Negative loan amounts
- Duplicate customer IDs
- Age values below 18
- Employment status entered differently (FT, Full Time, fulltime)
- Extreme outliers in income

Tasks:

1. Identify critical issues that could impact loan decisions. (5 Marks)

Answer:-

Problem	Example	Impact
Missing credit score	L003	Risk cannot be properly evaluated
Different income formats	900K vs 900000	Incorrect financial comparison
Negative loan amount	L004 = -200000	Invalid
Duplicate customer	L008 twice	Double counting
Age below 18	L005 = 16	Eligibility concern
Employment inconsistency	FT, Full Time, fulltime	Classification problem
Extreme income	L007 = 75,000,000	Possible outlier
Missing income	L010	Risk calculation affected

2. Propose cleaning techniques for missing and inconsistent data. (5 Marks)

Answer:-**Credit score**

- Check the original credit bureau data.
- If unavailable, retain NULL rather than inventing a score.
- For statistical analysis, imputation can be performed, but loan approval should not automatically rely on an imputed score.

Income

Convert:

900K → 900,000

750000 → 750,000

Use one standard unit, such as INR.

Loan amount

-200000 is invalid.

Verify whether the negative sign is a data-entry error. Do not automatically convert it to positive without verification.

Employment status

Standardize:

- FT
- Full Time
- fulltime

→ **Full Time**

Other categories can be standardized similarly.

Duplicate IDs

L008 occurs twice with identical data.

Keep only one record.

Age

Applicants below the allowed age should be flagged for eligibility review.

Missing income

Check the applicant's financial records. If unavailable, keep it missing and prevent the model from treating missing income as zero.

3. Detect and justify treatment of outliers. (5 Marks)

Answer:- L007 has annual income of **75,000,000**, which is extremely high compared with most other records.

Possible methods:

IQR method

Calculate:

$$\text{IQR} = Q3 - Q1$$

An observation can be considered an outlier if:

$$\text{Value} > Q3 + 1.5 \times \text{IQR}$$

or

$$\text{Value} < Q1 - 1.5 \times \text{IQR}$$

Z-score

For approximately normally distributed data:

$$Z = (x - \mu) / \sigma$$

Large absolute Z-scores can identify extreme observations.

However, an outlier should **not automatically be deleted**.

The bank should verify whether ₹75 million is genuinely the applicant's income. If valid, retain it. If it is a data-entry error, correct it.

4. Analyze the effect of cleaning on loan approval rates. (5 Marks)

Answer:-

Before cleaning:

- Approved = 7
- Rejected = 2
- Total = 9

Therefore:

$$\text{Approval rate} = 7/9 \times 100 = 77.78\%$$

After removing the duplicate L008, there are 8 unique applications.

If the duplicate is removed but all other records remain:

$$\text{Approved} = 7$$

$$\text{Total} = 8$$

$$\text{Approval rate} = 87.5\%$$

This apparent increase occurs because the duplicate record was an approved application.

However, this does **not** mean cleaning actually made the bank approve more loans. It simply removed an incorrectly duplicated record from the denominator.

If invalid or unresolved records are excluded, the approval rate may change again. Therefore, the bank should report both:

- Raw approval rate
- Cleaned approval rate
- Number of records removed/flagged
- Reason for each exclusion

5. Discuss ethical implications of incorrect data cleaning decisions.

Answer:- Incorrect cleaning can have serious consequences because loan approval directly affects people's financial opportunities.

Risks include:

- Unfair rejection of applicants
- Discrimination
- Incorrect credit-risk assessment
- Financial loss
- Lack of transparency
- Violation of regulatory requirements
- Bias caused by inappropriate imputation
- Removing legitimate outliers
- Favoring certain customer groups

Conclusion: Banking data cleaning must be accurate, explainable, auditable and fair. Cleaning rules should never be designed simply to produce a preferred approval rate.

Question 4: Smart City Traffic Monitoring Dataset

SensorID	Timestamp	VehicleCount	AverageSpeed	TrafficSignalStatus	RoadName	WeatherCondition
S101	01-01-2025 08:00	120	45	Green	Anna Salai	Sunny
S102	01-01-2025 08:05	135	52	Red	anna salai	Sunny
S103	01-01-2025 08:10	-50	48	Green	ANNA SALAI	Cloudy
S104	01-01-2025 08:15	140	350	Yellow	GST Road	Rainy
S105	NULL	110	42	Green	GST ROAD	Sunny
S106	01-01-2025 08:25	130	46	Red	OMR	NULL
S106	01-01-2025 08:25	130	46	Red	OMR	NULL
S107	01-01-2025 08:30	150	55	Green	Old Mahabalipuram Road	Sunny
S10A	01-01-2025 08:35	145	53	Green	OMR	Cloudy
S109	01-01-2025 08:40	138	50	Green	GST Road	Rainy

Data Issues:-

- Missing timestamps
- Duplicate sensor readings
- Vehicle counts recorded as negative numbers
- Average speed exceeding 300 km/h
- Missing weather information
- Inconsistent road naming conventions

Tasks:-

1. Assess the quality of the traffic dataset. (5 Marks)

Answer:-

Issue	Example	Classification
Missing timestamp	S105	Missing value
Duplicate reading	S106	Duplicate
Negative vehicle count	S103 = -50	Invalid value
Unrealistic speed	S104 = 350 km/h	Outlier
Missing weather	S106	Missing value

Issue	Example	Classification
Road-name variation	Anna Salai, anna salai, ANNA SALAI	Formatting
Road-name variation	GST Road, GST ROAD	Formatting
Long road-name variation	Old Mahabalipuram Road	Standardization
Invalid Sensor ID	S10A	Format error

2. Recommend cleaning techniques for time-series data. (5 Marks)

Answer:-

Timestamp validation

- Convert timestamps to a standard format.
- Example:

01-01-2025 08:00

→ 2025-01-01 08:00:00

Duplicate removal

Identify duplicate combinations of:

Sensor ID + Timestamp

S106 has the same reading twice, so one copy should be removed.

Missing timestamps

- Check neighboring readings.
- If the timestamp can be reliably determined from the sensor sequence, reconstruct it.
- Otherwise mark it missing.

Negative vehicle counts

Vehicle count cannot normally be negative.

Therefore:

-50 → invalid

Check the raw sensor logs before correction.

Missing weather

Use sensor station/weather-service information where appropriate, or retain NULL if the value cannot be recovered.

Road names

Standardize:

- Anna Salai
- anna salai
- ANNA SALAI

→ **Anna Salai**

and:

- GST Road
- GST ROAD

→ **GST Road**

3. Design rules to identify faulty sensor readings. (5 Marks)

Answer:-

The system can use validation rules such as:

Vehicle count

Vehicle Count ≥ 0

If:

Vehicle Count < 0

→ Flag as faulty.

Speed

Set a reasonable operational range based on the road/sensor characteristics.

For example:

Average Speed = 350 km/h

is clearly suspicious and should be flagged.

Timestamp

Ensure timestamps:

- Are valid
- Are chronological
- Do not contain impossible dates/times

Duplicate rule

If:

Sensor ID + Timestamp

is repeated with identical values, classify it as a duplicate.

Sensor ID

Sensor IDs should follow a predefined pattern such as:

S101, S102, S103...

An ID such as S10A should be flagged for verification.

4. Evaluate how cleaning affects congestion analysis. (5 Marks)

Answer:-

Traffic congestion analysis depends heavily on accurate vehicle counts and speeds.

The following errors can distort congestion:

- Negative vehicle count → artificially lowers traffic
- 350 km/h speed → artificially lowers congestion
- Duplicate readings → artificially increases traffic
- Missing readings → creates gaps in traffic patterns
- Inconsistent road names → splits the same road into multiple categories

For example, if Anna Salai, anna salai, and ANNA SALAI are treated as three roads, the system will underestimate the total traffic on Anna Salai.

After cleaning, traffic can be correctly aggregated by:

Road + Timestamp + Sensor

This produces more reliable congestion levels and traffic-signal decisions.

5. Suggest automated monitoring techniques for future data collection. (5 Marks)

Answer:-

5. Automated monitoring techniques

The city can implement:

1. Real-time validation rules
2. Range checks
3. Duplicate detection
4. Timestamp validation
5. Sensor heartbeat monitoring
6. Automated anomaly detection
7. Missing-data alerts
8. Sensor health dashboards
9. Automated road-name standardization
10. Data-quality scores
11. Alerting when abnormal speeds/counts occur
12. Historical comparison of sensor readings

Conclusion: Automated monitoring allows faulty sensor readings to be detected before they affect traffic management.

Question 5: Global Sales Performance Dataset

Region	Country	SalesPerson	Product	SalesAmount	Currency	SalesDate	CustomerRating
Asia	India	Raj Kumar	Laptop	50000	INR	10-01-2025	4
Asia	India	Rajkumar	Laptop	55000	INR	10-01-2025	5
Europe	Germany	Anna M	Laptop	1200	EUR	12-01-2025	4
Europe	Germany	Anna M	Laptop	1200	EUR	12-01-2025	4
North America	USA	John D	Laptop	1500	USD	01-15-2025	6
Asia	India	Priya S	Mobile	NULL	INR	18-01-2025	3
Europe	France	Pierre L	Ordinateur Portable	1300	EUR	20-01-2025	4
Asia	India	Rahul P	Mobile	9999999	INR	25-01-2025	5
UK	UK	David T	Laptop	1100	GBP	28-01-2025	2
Asia	India	Priya S	Smartphone	35000	INR	30-01-2025	0

Data Issues

- Multiple currencies (USD, EUR, INR, GBP)
- Missing sales amounts
- Duplicate regional entries
- Customer ratings outside the scale (1–5)
- Different date formats across countries
- Product names in multiple languages
- Salespersons recorded with spelling variations
- Extreme sales outliers due to data-entry errors

Tasks:-

1. Develop a comprehensive data-cleaning framework for multinational datasets. (5 Marks)

Answer:- **Step 1 — Data profiling**

Identify:

- Missing values
- Duplicates
- Outliers
- Invalid values
- Different formats

Step 2 — Standardization

Standardize:

- Currency
- Dates
- Product names
- Salesperson names
- Regions
- Countries

Step 3 — Validation

Apply rules such as:

Customer Rating $\in [1,5]$

Sales Amount ≥ 0

Sales Date must be valid

Step 4 — Duplicate detection

Compare:

- Region
- Country
- Salesperson
- Product
- Sales Amount
- Sales Date

Step 5 — Missing-value treatment

Investigate the original source first.

For example:

Priya S — Mobile — NULL

should be checked against the original sales system.

Step 6 — Outlier detection

Use:

- IQR
- Z-score
- Historical sales comparison
- Business rules

Step 7 — Audit

Maintain:

- Original value
- Cleaned value
- Reason for modification
- Date of modification
- Person/system responsible

2. Design methods to standardize currencies, products, and dates. (5 Marks)

Answer:-

Currency

The dataset contains:

- INR
- USD
- EUR
- GBP

Convert all amounts to a common reporting currency.

For example:

Standard Sales Amount = Original Sales Amount × Historical Exchange Rate

The original currency should still be preserved.

The assessment does not provide exchange rates, so exact converted totals cannot be calculated from the supplied data.

Product names

Examples:

Laptop → Laptop

Ordinateur Portable → Laptop

Mobile → Mobile

Smartphone → Smartphone

A master product dictionary should be maintained.

Dates

Convert:

10-01-2025

12-01-2025

01-15-2025

into one standard format:

YYYY-MM-DD

However, dates such as 01-15-2025 need locale-aware interpretation because the month/day order differs from some other records.

3. Handle missing values and outliers using appropriate analytical techniques. (5 Marks)

Answer:- **Missing sales amount**

For Priya S's Mobile transaction:

Sales Amount = NULL

Recommended approach:

1. Check the original sales system.
2. Recover the actual value if possible.
3. If unavailable, keep it missing.
4. For statistical analysis, use an appropriately documented imputation method.
5. Flag the value as imputed.

Customer ratings

Ratings must be between 1 and 5.

Therefore:

- Rating 6 → invalid
- Rating 0 → invalid

These should be verified or changed to NULL until the original values are confirmed.

Extreme sales outlier

Rahul P has:

₹9,999,999

This is dramatically larger than the other Indian sales values and may be a data-entry error.

It should be investigated rather than automatically deleted.

Possible techniques:

- IQR
- Z-score
- Historical customer sales
- Product price validation
- Manual verification

4. Compare regional sales rankings before and after cleaning. (5 Marks)

Answer:- **Before cleaning**

Using the raw numeric amounts as displayed, without currency conversion:

Region	Raw Sales
Asia	10,139,999
Europe	3,700
North America	1,500
UK	1,100

Therefore, the raw ranking is:

1. Asia
2. Europe
3. North America
4. UK

However, this ranking is misleading because different currencies are being added together.

Major cleaning changes

The Asia total is dominated by the apparent outlier:

₹9,999,999

Removing that suspected error and removing duplicate/duplicate-like entries gives a much more realistic picture.

For example, the clearly duplicated European Anna M transaction should be counted once.

The cleaned raw INR values for the clearly valid Indian entries are:

- Raj Kumar Laptop = ₹50,000
- Rajkumar Laptop = ₹55,000
- Priya S Smartphone = ₹35,000

Total:

₹140,000

The missing Priya Mobile transaction remains unresolved.

Therefore, cleaning dramatically reduces the apparent dominance of Asia.

However, an exact final regional ranking **after cleaning cannot be reliably calculated** without:

- Exchange rates
- Verification of the ₹9,999,999 transaction
- Missing sales amount
- Confirmation of duplicate/identity matches

This is important because the assessment specifically requires currency standardization and comparison of regional rankings.

5. Prepare an executive report explaining how data quality influences strategic business decisions.

Answer:-

Executive Summary

Data quality has a direct effect on strategic business decisions. The global sales dataset contains multiple currencies, missing values, duplicate transactions, invalid ratings, inconsistent dates, multilingual product names, salesperson spelling variations and extreme outliers.

If these problems are not corrected, management may incorrectly identify the best-performing regions, products and salespersons.

For example, the ₹9,999,999 sales value causes Asia's raw sales total to become extremely large. If this value is a data-entry error, management could incorrectly conclude that Asia is dramatically outperforming other regions.

Duplicate records can also inflate sales figures, while missing sales amounts can underestimate revenue.

Incorrect customer ratings may affect customer satisfaction analysis, and inconsistent product names can split the same product into multiple categories.

Business impact

Poor data quality can result in:

- Incorrect regional rankings

- Incorrect sales forecasts
- Wrong inventory decisions
- Incorrect employee performance evaluation
- Financial reporting errors
- Poor marketing decisions
- Incorrect product strategies
- Loss of management confidence in analytics

Recommendations

The company should establish:

1. A centralized data dictionary
2. Standard currency conversion rules
3. Master product and salesperson databases
4. Automated validation
5. Duplicate detection
6. Outlier monitoring
7. Mandatory fields
8. Data-quality dashboards
9. Audit trails
10. Regular data-quality reviews
11. Staff training
12. Clear data-governance ownership

Conclusion: High-quality data is essential for reliable analytics and strategic decision-making. Data should be validated, standardized and monitored continuously before being used for reporting, forecasting or business decisions.